

Graph neural networks for clinical risk prediction based on electronic health records: A survey

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ABSTRACT

Objective: This study aims to comprehensively review the use of graph neural networks (GNNs) for clinical risk prediction based on electronic health records (EHRs). The primary goal is to provide an overview of the state-of-the-art of this subject, highlighting ongoing research efforts and identifying existing challenges in developing effective GNNs for improved prediction of clinical risks.

Methods: A search was conducted in the Scopus, PubMed, ACM Digital Library, and Embase databases to identify relevant English-language papers that used GNNs for clinical risk prediction based on EHR data. The study includes original research papers published between January 2009 and May 2023.

Results: Following the initial screening process, 50 articles were included in the data collection. A significant increase in publications from 2020 was observed, with most selected papers focusing on diagnosis prediction ($n = 36$). The study revealed that the graph attention network (GAT) ($n = 19$) was the most prevalent architecture, and MIMIC-III ($n = 23$) was the most common data resource.

Conclusion: GNNs are relevant tools for predicting clinical risk by accounting for the relational aspects among medical events and entities and managing large volumes of EHR data. Future studies in this area may address challenges such as EHR data heterogeneity, multimodality, and model interpretability, aiming to develop more holistic GNN models that can produce more accurate predictions, be effectively implemented in clinical settings, and ultimately improve patient care.

1. Introduction

Electronic health records (EHRs) are extensive, heterogeneous, and longitudinal repositories that document patients' health, including symptoms, prescriptions, clinical notes, and medical images. With the increase in EHR data collection, there is growing interest in leveraging this information to improve patient care, especially in the context of clinical risk prediction [1]. Recent machine learning approaches focused on predicting events such as disease diagnoses, mortality, and hospital readmissions have been relevant to this endeavor [2,3].

Despite the rich information present in EHRs, translating it into actionable insights presents challenges due to data-related problems such as heterogeneity (multiple types of medical attributes describing

a patient), high dimensionality (a large number of attributes associated with a patient), quality (missing values and inconsistencies) and temporal dynamics (numerous patient encounters and timestamped clinical events) [1,4–7]. Considering that the success of machine learning models depends largely on an adequate representation of the input data, studies in representation learning – the process of learning expressive representations of the input data for improved performance of predictors [8] – are paramount for effectively transforming patient data from the raw EHR format into adequate representations that fully capture their health status [1].

Recent deep learning techniques have effectively addressed these challenges. Unlike traditional machine learning approaches, which rely

Abbreviations: EHR, Electronic health record; GNN, Graph neural network; GCN, Graph convolutional network; GAT, Graph attention network; GAE, Graph autoencoder; CNN, Convolutional neural network; RNN, Recurrent neural network; GRU, Gated recurrent unit; LSTM, Long short-term memory

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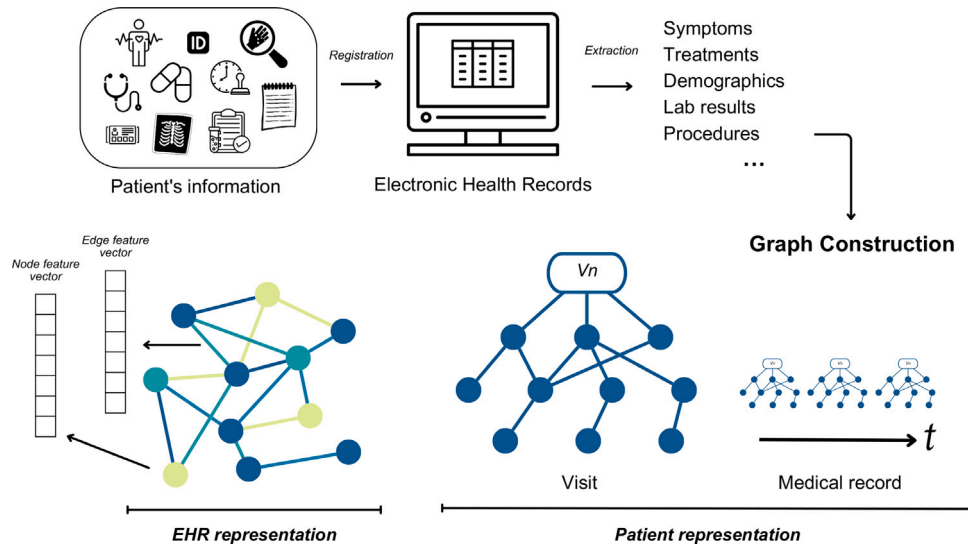


Fig. 1. Electronic health records contain a range of multimodal patient data. This information can be used for patient graph representations, focusing on a patient's visit or medical record. In the case of a visit, a hierarchical and homogeneous graph is shown, while medical records are made up of sequences of visit graphs. Alternatively, the entire EHR data can be modeled as a heterogeneous graph, with different types of nodes and edges represented by different colors. In all examples, nodes and edges can have feature vectors processed using GNN for further clinical risk prediction tasks. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

heavily on expert-driven feature engineering, deep learning models can automatically extract meaningful latent feature representations from complex raw data [9–11]. Among these, graph neural networks (GNNs) stand out. The goal of graph representation learning is to encode graphs into a low-dimensional vector space while preserving topology and node properties [12]. In this sense, GNNs are particularly adept at representing EHRs because they can capture the intricate relationships and dependencies between medical entities to generate rich, context-aware embeddings for further downstream tasks [13,14]. This is a promising feature in contrast to other machine learning and deep learning algorithms, which often treat medical concepts as a flat “bag of features”, disregarding structural information and variable interdependencies during model development [15,16]. Furthermore, GNNs are powerful in handling the high sparsity and frequent missing values found in EHR data, as they can respectively propagate information through the graph structure to densify representation and infer features based on the attributes of neighboring nodes in the medical graph [17–19].

The strength of GNNs lies in their capability to navigate the intricacies of non-Euclidean spaces [13,20]. Unlike grid-based data structures such as images, which have inherent locality and consistent neighboring relationships, graphs often lack a natural node ordering, and the spatial proximity of nodes does not determine their relationships, making it more challenging to apply key operations such as convolutions [21,22]. For example, EHR graphs can represent a dense web of patient histories, diagnoses, treatments, and other clinical outcomes, with heterogeneity in node types, nodes with varied degrees, and edges indicating co-occurrence, causality, hierarchical relations, and other relevant interactions, resulting in complex topological structures (Fig. 1). In this sense, GNNs offer the necessary flexibility to capture and exploit these relationships, yielding thorough representations and, consequently, improved interpretability and efficiency compared to other deep learning models [23].

Early studies on GNNs for clinical risk prediction aimed to take advantage of hierarchical medical information, structured as ontologies and knowledge graphs, as distant supervision [24]. They introduced label information through structured knowledge graph propagation, learning correlations between medical codes and paralleling them with codes observed in patients to obtain better predictions [24]. This approach enabled more accurate predictions than other deep learning

baseline models [25]. Subsequent approaches started to prioritize the learning of novel graph representations based on EHR rather than the integration of knowledge graphs. Some of these representations include patient similarity, patient-medication interactions, and temporal relations between medical events [26–28]. Today, given the plethora of existing EHR multimodal information, heterogeneous graphs have also been used, including clinical notes, disease codes, medical images, and lab results into the learned embeddings, enriching the representations of the data used for critical health predictions [29,30].

The manifold use of GNNs in EHRs represents a transformative paradigm in the landscape of clinical task predictions. GNNs, as powerful tools for modeling complex relationships within graph-structured data, have demonstrated remarkable efficacy in capturing intricate dependencies inherent in healthcare systems and will likely support future disruptive advances in this domain. Notably, recent studies have discussed the applications of deep learning in electronic health records. However, none have explicitly focused on using GNNs for clinical risk prediction based on EHR. For example, [7] have concentrated on temporal patient presentation, while [1,11] have focused on general deep learning techniques for EHR. The most recent studies concerning graph representation have limited their scope to diagnosis prediction only [31] or did not focus on GNNs [32].

Thus, while preceding review papers have explored the broader landscape of deep learning applications in EHRs, a dedicated review addressing the intricacies of utilizing GNNs for clinical risk prediction based on EHRs remains an unexplored niche in the literature. The narrative review presented in this paper aims to bridge this gap by offering a targeted exploration of the advancements, open challenges, and potential future research directions in this specific application domain. Inspired by systematic protocols, the intention is to summarize the current scope and depth of the available literature, while also setting the stage for future systematic reviews as the field grows, providing a fundamental overview essential for advancing research in the area (see Table 1).

2. Background

2.1. EHR representation learning

The primary aim of representation learning for EHRs is to transform input data into a suitable representation that can enhance downstream

Table 1
Significance of the presented survey.

Problem.	The abundance of heterogeneous EHR data and the relations among medical events and entities are two key factors that pose challenges for developing clinical risk prediction models.
What is already known.	Leveraging the relational nature of medical data as graphs with deep learning approaches can improve clinical risk prediction.
What this paper adds.	A narrative review of the state-of-the-art GNN techniques for EHR-based clinical risk prediction. The study highlights the main challenges related to EHR graph data representation, temporality, and prediction tasks and describes various GNN architectures used in this endeavor.

clinical prediction tasks [15]. Early studies on EHR representation learning, exemplified by Doctor AI [33], RETAIN [34], and Dipole [35], utilized recurrent neural network (RNN)-based models to account for the sequential nature of EHR data in clinical risk prediction. Convolutional neural networks (CNNs), word embedding methods, and stacked denoising autoencoders were also utilized [36–42]. These methodologies aimed to model dependencies between medical codes of the International Classification of Diseases (ICD), an international standard for diagnosing and classifying health conditions [43].

Subsequent studies recognized the need for integrating structured information into models, emphasizing the increasing relevance of graph representation learning approaches. In the Heterogeneous Convolutional Neural Network (HCNN), a CNN was adapted to capture the temporal relationships between medical events in EHR data, which were represented as attributed graphs [44]. Attention-based models such as GRAM [45], KAME [46], MMORE [47], and CAMP [48] also incorporated hierarchical diagnosis information contained in the ICD ontology for clinical risk prediction. The MiME model employed multilevel relationships to learn representations of medical concepts [49]. Other similar studies that have aimed to model EHR as graphs include [50–53].

2.2. Graph neural networks

Graph neural networks (GNNs) are deep learning models designed for processing and analyzing data organized as graphs [54]. A graph can represent any relationships (edges) between a collection of entities (nodes) [55]. Graphs are a fundamental way in which it is possible to obtain data from the natural world, and a large part of patterns observed in nature can be expressed and understood using graph structures [56]. In this way, GNNs are a powerful resource for representation learning, as they can distill structural information and learn powerful high-level representations [57]. GNNs have demonstrated applicability across many network-related areas, including traffic forecasting, social network analysis, and improved recommendation systems [56]. In healthcare and biology, they have been used to discover new drugs, predict protein–protein interactions, forecast disease outbreaks, and assess patient similarity [15,58–63].

A graph is conventionally defined as $G = (V, E)$, where V is the set of nodes, and E is the set of edges that connect pairs of nodes in V [56]. In EHRs, depending on the approach, nodes can represent various healthcare entities such as patients, diagnoses, and medications (Fig. 1). Each node $u \in V$ has an associated feature vector $x_u \in \mathbb{R}^k$. These features are organized into a node feature matrix $X \in \mathbb{R}^{|V| \times k}$, where each row denotes a node's feature, making them suitable for machine learning models [56]. For instance, a feature vector for a patient node can include encoded attributes such as age, gender, medical history information, and vital signs. Furthermore, an edge set E is typically represented by an adjacency matrix $A \in \mathbb{R}^{|V| \times |V|}$,

indicating whether node u is related to another node v through binary indicators ($a_{uv} = 1$ or 0). The presence of an edge between a patient node and a medication node could indicate, for example, whether a patient received specific medication during a visit. Nonetheless, this is a simplified representation where edges are not attributed, meaning they do not encapsulate additional information, such as the strength or type of relationship. Even in such cases, the core properties of the GNNs remain the same.

GNNs operate by aggregating information from a node's immediate neighborhood, which is then combined with the node's features to create a richer latent representation (Fig. 2) [64,65]. This is achieved with matrix multiplication of the adjacency (A) and feature (X) matrices [66]. After k aggregation steps, the structural information within a node's k -hop neighborhood is captured, and a final graph representation can be obtained by pooling the transformed feature vectors of all nodes [67]. This propagation mechanism, often referred to as message passing, enables GNNs to learn expressive representations and solve tasks such as node classification (predicting the label of a node), link prediction (predicting whether there is an edge between two nodes), and graph classification (predicting a label for the entire graph) [68].

GNNs must follow the principles of permutation invariance and equivariance to handle graph structures. This ensures that their output remains consistent, regardless of how nodes are presented [56]. Furthermore, GNNs must handle varying numbers of node neighbors. For instance, one patient node might be linked to several medication nodes during a visit, while another patient might be connected to only a few, and the same model must address both cases. This is achieved by employing local functions (f) that compute node-level outputs (h_u) by considering not only the features of the target node (u) but also those of its specific number of neighboring nodes [56]. The choice of particular local functions depends on the GNN architecture; these variations are further discussed in the following sections.

Note that this study focuses on GNNs utilized for EHR data analysis. For a comprehensive review encompassing various types of GNNs, interested readers are referred to [54–56,67–69].

2.2.1. Graph convolutional networks

Graph convolutional networks (GCNs), introduced by Kipf and Welling [70], propagate information across the graph and aggregate it to update node representations, extending convolutions from the Euclidean domain to the graph domain — which is characterized by data structured as nodes and edges [20]. In the spatial approach, each GCN layer computes new node representations based on their current features and those of their neighbors, similar to how convolutional layers in CNNs aggregate local information in images — but in a configuration where operation objects are non-fixed in size [54,71]. As nodes progress through the GCN layers, the learned representations encapsulate a broader neighborhood, similar to the increasing receptive fields in CNNs. The local function that provides the update rule for node v in a GCN involves a weighted sum of the features of the node and its neighbors and is given by

$$h_v^{(l+1)} = \sigma \left(\sum_{u \in N(v) \cup \{v\}} \frac{1}{c_{vu}} W^{(l)} h_u^{(l)} \right)$$

where $h_v^{(l)}$ is the feature vector of node v in layer l , $N(v) \cup \{v\}$ is the set of neighbors of v and v itself, c_{vu} is a normalization constant, $W^{(l)}$ is a weight matrix, and σ is a nonlinear activation function.

GCNs can also be developed using spectral approaches [72]. Spectral models treat graphs as signals processed through graph convolution in the spectral domain. In particular, graph signals are first transformed into the spectral domain using a Graph Fourier Transform (GFT), which leverages the eigenvectors of the graph Laplacian as its basis functions. Specifically, given a graph with the Laplacian matrix L , its eigenvalue decomposition leads to eigenvectors that serve as harmonic modes for the GFT. Once in the spectral domain, the graph signals are filtered

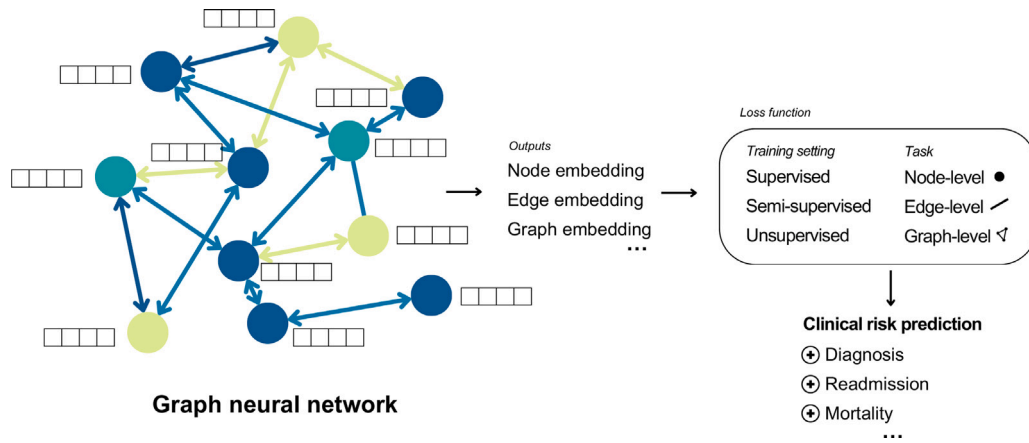


Fig. 2. Simple representation of how a GNN layer operates for clinical risk prediction. The double-edge arrows represent the message-passing process between neighboring nodes. This interaction allows nodes to aggregate information and generate context-aware embeddings. The model's design, including its outputs and loss functions, are adapted based on the specific requirements of the tasks.

and subsequently transformed back to the spatial domain with the GFT [54].

Despite their different starting points, both spectral and spatial models iteratively collect neighborhood information (organized as low-dimensional vectors) to capture high-order correlations among the analyzed graphs [57].

2.2.2. Graph attention networks

Graph attention networks (GATs), proposed by Veličković et al. in 2018, extend GCNs by introducing an attention mechanism to weigh the importance of neighboring nodes [59]. This mechanism is inspired by the attention of the Transformer model, which allows the network to simultaneously focus on different parts of the input [73].

The local function that provides the update rule for GATs involves two main steps: calculating attention scores for each pair of nodes in the graph, which indicate the relevance of neighbor nodes' features to a given central node; and a weighted aggregation of the features of the central node and its neighbors, where the previously calculated attention scores determine the weights [59]. The rule is then defined by

$$h_i^{(l+1)} = \sigma \left(\sum_{j \in N(i)} \alpha_{ij} W h_j^{(l)} \right)$$

where α_{ij} is the attention coefficient computed as

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(a^T [W h_i \parallel W h_j]))}{\sum_{k \in N(i)} \exp(\text{LeakyReLU}(a^T [W h_i \parallel W h_k]))}$$

GATs offer particular value in interpretability, as learned attention weights enable a deeper understanding of the relevance of specific nodes for a given end task [74].

2.2.3. Graph autoencoders

Graph autoencoders (GAEs) are unsupervised learning models that aim to learn compact representations of graph nodes, acting as a type of dimensionality reduction technique [75]. The local function in GAEs is employed in encoding and decoding graph information. During encoding, node features are transformed into a lower-dimensional latent representation, capturing information both about its neighboring nodes and the general graph structure. During decoding, the latent representations are used to reconstruct the original features while minimizing the reconstruction error [76].

In 2016, Kipf and Welling extended the GAE framework by introducing a variational inference approach called variational graph autoencoder (VGAE) [77]. In VGAE, the encoder not only generates a low-dimensional vector representation for each node but also models

the underlying probability distribution of these representations, usually with a Gaussian distribution. The decoder, then, aims to minimize the difference between the distributions of the reconstructed data and the actual data, allowing the model to account for uncertainty and leading to more robust graph reconstructions.

3. Methods

3.1. Search strategy and information sources

A comprehensive, narrative review of the use of GNNs in EHR-based clinical risk prediction was conducted. The PRISMA protocol (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) [78] was used as inspiration for the adopted review process, as detailed in Fig. 3. However, the option was for a more adaptable strategy to provide a comprehensive summary and perspective on GNNs, which led to qualifying the review as “narrative” rather than “systematic”. This decision reflects the intention to offer a broad, interpretive overview of the literature, emphasizing context and insight rather than a narrower quantitative analysis, which we believe is particularly valuable in the developing field of GNNs for clinical risk prediction. Five databases were used: Scopus, PubMed, ACM Digital Library, and Embase. The search term used was as follows:

“Graph” OR “Graph neural network” OR “Graph neural” OR “GNN” OR “Graph convolutional” OR “GCN” OR “Graph autoencoder” OR “GAE” OR “Graph attention” OR “Graph attention network” OR “GAT” OR “Graph self-attention” OR “GSA” OR “Graph transformer” OR “Graph transformer network” OR “GTN” OR “Graph transformer” OR “Graph-based” OR “Graph embedding”) AND (“Electronic health record” OR “EHR” OR “Electronic medical record” OR “EMR” OR “electronic health data”) AND (“Deep learning” OR “Neural network” OR “Representation learning” OR “Artificial intelligence”)

An initial assessment of the literature using these terms revealed a large number of potentially relevant articles. Therefore, to maintain specificity, we chose not to include the more general terms “machine learning” and “clinical”, which could dilute the focus of the analysis. The gray literature, specifically the ArXiv repository, was also revised using similar terms. The search was limited to articles published between January 1, 2009, and May 14, 2023, as it was anticipated that there would be no relevant articles published before 2009 since GNNs started to become more popular after Scarselli et al. (2009) [65].

3.2. Eligibility criteria

This review aimed to include all related articles published in English that used GNNs for clinical risk prediction based on EHRs within the search period. The exclusion criteria employed during the screening

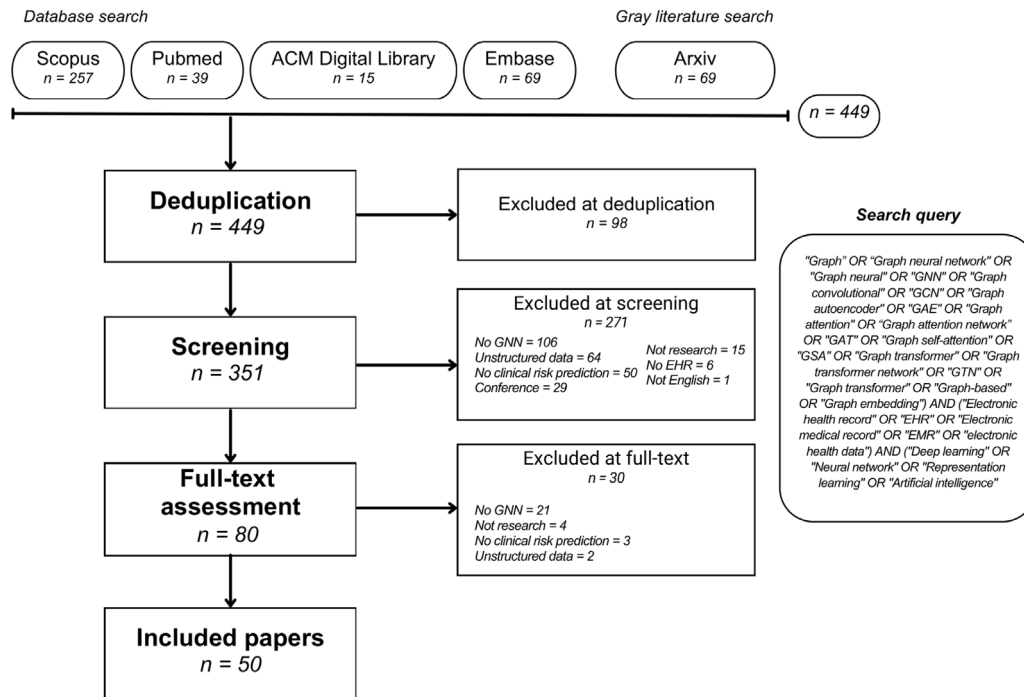


Fig. 3. Flowchart for literature search and selection of articles for the review.

and full-text review stages encompassed the following attributes: utilization of unstructured EHR data as the main focus; absence of clinical risk prediction focus; non-research or conference content; and paper not in the English language. Review articles, duplicate records, and studies that did not include EHRs or GNNs were also excluded. Three authors screened the abstracts, while the first author reviewed the full texts and extracted the data. Data extraction results were revised by the other authors. The search strategy and selection process are summarized in Fig. 3.

3.3. Data extraction, synthesis, and analysis

First, the selected papers were categorized based on the specific clinical risk prediction task(s) they intended to solve. Then, details were gathered about the task prediction levels (patient or visit), graph representation, and node definitions, including whether they incorporated temporal information and the type of resource employed to help in the learning process, if any. Concerning the model's implementation, data were extracted regarding the EHR datasets used, patient counts, evaluation metrics implemented in each study, available repository links, and techniques utilized for interpretability, if any. Finally, all the collected information was consolidated for subsequent qualitative and quantitative analyses, centered on three axes: data representation, incorporation of temporality, and prediction tasks.

3.4. Summary of study selection

After deduplication of the initial 449 articles, 351 papers were obtained. Out of these, 50 articles were selected for full-text review and data collection following the steps detailed in Fig. 3. Among the 69 preprints retrieved from Arxiv, only one study was eligible for the review and included in the selected articles. Table 2 summarizes the statistics of the 50 papers included in this review. Detailed information for all papers is provided in Supplementary Table, and an overview of the results is provided in the next sections.

4. Results

The Results section is structured as follows: first, a brief summary of the main results is presented. Next, a more detailed description of

the main graph data representation approaches is provided, including the incorporation of temporality into clinical predictions, and the most common clinical risk prediction tasks. Finally, a discussion addresses the unique challenges of modeling EHR data with GNNs, presenting perspectives in the field, and describing study limitations.

4.1. Overview of the studies characteristics

Using GNNs for clinical risk prediction is a rising trend, as shown in Fig. 4. There were numerous approaches to graph representation in the evaluated articles, which often overlap. One of the main ones involves using a hierarchical graph based on codes from medical ontologies, such as ICD (diseases, procedures) and ATC (medications), which describe well-known parent-child relations ($n = 15$, Table 3). In this case, the hierarchical medical code graph is considered homogeneous. Another technique involves learning heterogeneous EHR graphs, where nodes other than medical codes are comprised, such as patients, lab values, and doctors, and edge relations may vary ($n = 9$, Table 3). These two approaches are often combined to create a heterogeneous graph where medical ontology knowledge is also an aspect integrated into the learned embeddings ($n = 5$, Table 3). Other techniques include similarity networks, where nodes are linked based on feature similarities ($n = 6$); bipartite graphs, with two types of nodes ($n = 5$); hypergraphs, where edges can connect any number of nodes ($n = 2$); dynamic graphs, where features change over time; and multi-view graphs, which include a combination of different types of networks of medical entities interactions ($n = 3$). For a summary, please refer to Table 3; more information can be found in the Supplementary Material.

Out of all the clinical risk prediction tasks, diagnosis prediction has received the most attention (Fig. 5, $n = 36$), followed by mortality and readmission. The diagnosis task also tested the most diverse GNN model architecture set (Fig. 6). An overview of GNNs used for these three main tasks is provided in Section 4.4.

When considering model architectures, it was possible to observe that graph attention networks (GAT) were the most employed ($n = 19$), followed by graph convolutional networks (GCN) ($n = 18$) and GraphSAGE ($n = 4$) (Fig. 7). Different architectures were also employed to handle unique medical data requirements, such as GRU, LSTM, and

Table 2

A summary of the selected articles.

Year	Ref.	Clinical risk prediction task				Data source ^a	Interp. ^b	Metrics				Reprod.
		Diag.	Mort.	Read.	Otr.			ACC	AUPRC	AUROC	F1	
2020	[16]	x	x	x		EI, SY	AW	0.77	0.53; 0.59			x
2020	[25]	x				M3		0.65				x
2020	[28]	x				M3, PV		0.94				
2020	[79]	x				M3, UT		0.48		0.72		
2020	[80]	x				M3, PV		0.92				
2020	[81]	x				PC		0.91	0.09	0.81	0.15	x
2020	[82]	x				M3		0.57				
2020	[83]		x			M3			0.34	0.9		
2020	[29]	x				M3	TS					x
2020	[84]	x				M3	TS	0.63	0.67	0.68		
2021	[18]	x	x	x		M3, EI, NY	SVA		0.71; 0.39			x
2021	[26]		x		x	EI	AW		0.43	0.85		x
2021	[85]	x		x		IQ, EI	PCA		0.82; 0.2		0.86; 0.67	x
2021	[30]	x				PV	AW			0.74	0.66	x
2021	[17]	x				PV					0.45	x
2021	[86]	x				M3	TS, H	0.60				
2021	[87]				x	M3, WH, SP, MB	AW	0.82		0.93		x
2021	[88]	x				PV						
2021	[89]			x		M3			0.84; 0.67	0.85; 0.67		x
2021	[90]	x				IB	U, C	0.79				x
2021	[91]	x				PV					0.53; 0.62	
2021	[92]			x		M3		0.37		0.78	0.42	
2021	[93]	x				M3				0.85	0.72; 0.48	x
2021	[94]	x			x	SY		0.9				
2022	[95]	x				M4, EI	AW	0.96; 0.15	0.60; 0.49	0.89; 0.8		x
2022	[96]	x				M3	TS				0.21	
2022	[97]	x				PV	AW				0.44	
2022	[98]	x				EM		0.75				
2022	[99]	x				CN			0.72	0.15		
2022	[100]	x				M3	TS	0.87			0.84	
2022	[101]	x	x			M3, XM			0.53; 0.85	0.87; 0.90	0.5; 0.84	
2022	[102]	x				M3, M4				0.9	0.73; 0.26	x
2022	[103]				x	PV	TS, GE			0.91		
2022	[104]		x		x	M3	TS		0.62; 0.65	0.90; 0.75		x
2022	[105]				x	NC	CIC		0.15	0.75		x
2022	[106]	x				CD						
2022	[107]	x				PH	C				0.15	x
2022	[108]	x				PV					0.81	
2022	[109]		x	x		M3, EI			0.62; 0.39	0.91; 0.72	0.53; 0.37	
2022	[110]	x	x			M3, CK, TJ, HM			0.76; 0.8; 0.54	0.83; 0.95; 0.86		x
2022	[111]		x			M4				0.68	0.53	
2022	[112]	x				M3, M4	TS	0.93				x
2023	[19]	x				CF	C			0.89	0.8	
2023	[113]				x	CU, UV	AW		0.88	0.95		
2023	[27]				x	PV				0.83	0.27	
2023	[114]	x				M4, PE				0.93	0.87; 0.89	
2023	[115]			x		M4, PV	GE			0.78		x
2023	[31]	x				M3, EI	TS, AW			0.86	0.64; 0.74	
2023	[116]	x				MC	S			0.99		x
2023	[12]				x	MC	TS					

^a EI: eICU, M3: MIMIC-III, M4: MIMIC-IV, PV: Private, UT: UTP, PC: Physionet Challenge, NY: NYU Langone Health, IQ: IQVIA US, WH: WHAS, SP: SUPPORT, MB: METABRIC, IB: IBM Explorys, SY: Synthetic, EM: EMRNet, CN: Cardionet, XM: Xiangya Medical, NC: N3C, CD: COPD, PH: PopHR, CK: CKD, CG: Cardiology, TJ: TJH, HM: HMH, CF: Cerner's Health Facts, CU: CNUH, UV: UV, PE: P-EHRs, MC: Mayo Clinic.

^b TS: t-SNE, A: Attention, SV: SVA, P: PCA, U: UMAP, GE: GNN Explainer, CIC: Clinically intuitive concepts, S: SHAP, EGE: Extended GNNExplainer, C: Clustering, AW: Attention weights.

Table 3

Types of EHR graph representation employed in the analyzed papers.

Graph representation	Papers
Hierarchical	[25,27,28,30,31,80,91–93,96,103,107, 108,112]
Heterogeneous	[29,88,95,97,100,110,111,114,116]
Hierarchical and heterogeneous	[16,19,82,105,106]
Similarity	[26,89,94,101,104,115]
Bipartite	[17,85,90,93,99]
Dynamic	[79,83,102]
Hypergraph	[96,109]
Other	[12,18,81,84,86,87,98,113]

BERT (Supplementary Table). In these cases, GNNs were responsible for the graph representation learning step, while the others were used for processing temporal aspects and multimodal data.

Regarding incorporating temporal patient information, most evaluated approaches consider some sort of clinical event sequentiality, such as the order of patient visits (Supplementary Table, $n = 32$). However, most of these works disregard irregular time intervals. More information about the temporal aspects of reviewed models is described in Section 5.5.

GNNs were evaluated against and consistently outperformed other machine learning and deep learning techniques in all evaluated studies, indicating that graph representations are valuable for clinical risk prediction. A summary of metrics is shown in Table 2, and a full report can be found in the Supplementary Table. Most authors compared the proposed GNN models with traditional and baseline approaches using

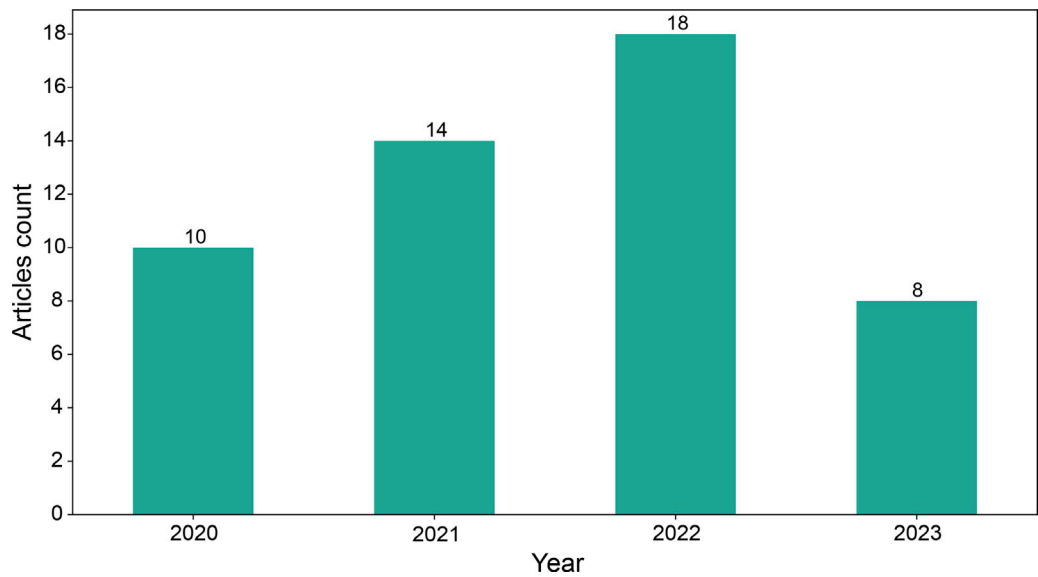


Fig. 4. Yearly distribution of selected articles. The cutoff date for including articles was May 14, 2023.

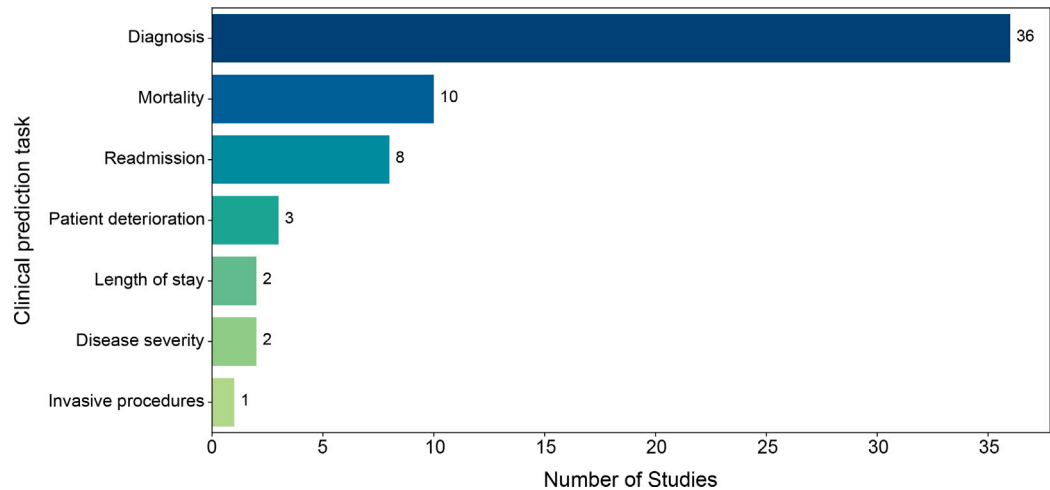


Fig. 5. Distribution of articles addressing different clinical risk prediction tasks.

Table 4
A summary of EHR data resources used in the evaluated articles.

Data resources	Papers
M3 (MIMIC-III)	[18,25,28,29,31,79,80,82–84,86,87,89,92,93,96,100–102,104,109,110,112]
M4 (MIMIC-IV)	[95,102,111,112,114,115]
EI (eICU)	[16,18,26,31,85,95,109]
SY (Synthetic)	[16,94]
Other	[12,17–19,27,28,30,79–81,85,87,88,90,91,97–99,101,103,105–108,110,113–116]

classic ML metrics such as Accuracy, Precision, Recall, and F1-score, as well as AUPRC (Area Under the Precision–Recall Curve) and AUROC (Area Under the Receiver Operating Curve).

The availability of resources and documentation highlights the aspect of reproducibility in the evaluated models. Of the 50 selected articles, 22 included a link to a repository containing the project’s

source code, as shown in Table 2 (links in the Supplementary Material). Most papers provided comprehensive details about their model architecture and parameters, including the specifics of GNN layers, frameworks, learning rates, optimizers, and data split percentages.

Furthermore, the use of benchmark datasets also helped with reproducibility efforts, as it allows for validations against a known standard. For example, the Medical Information Mart for Intensive Care III, or MIMIC-III [117], was the most frequently used dataset (n = 23, Table 2). MIMIC is a freely accessible database, one of the most widely employed EHR datasets, encompassing data from over 40,000 patients admitted to the Beth Israel Deaconess Medical Center in the USA, spanning the years 2001 to 2012. Other datasets include the eICU dataset [118], which provides data of over 200,000 admissions to various intensive care units (ICUs), MIMIC-IV [119], an updated and more extensive version of MIMIC-III, as well as synthetic EHR datasets, which are created based on the real structure of EHR datasets. Aside from these, a number of studies make use of proprietary and private datasets obtained directly from hospitals for their GNN analyses. A summary of utilized datasets and the corresponding articles is presented in Table 4.

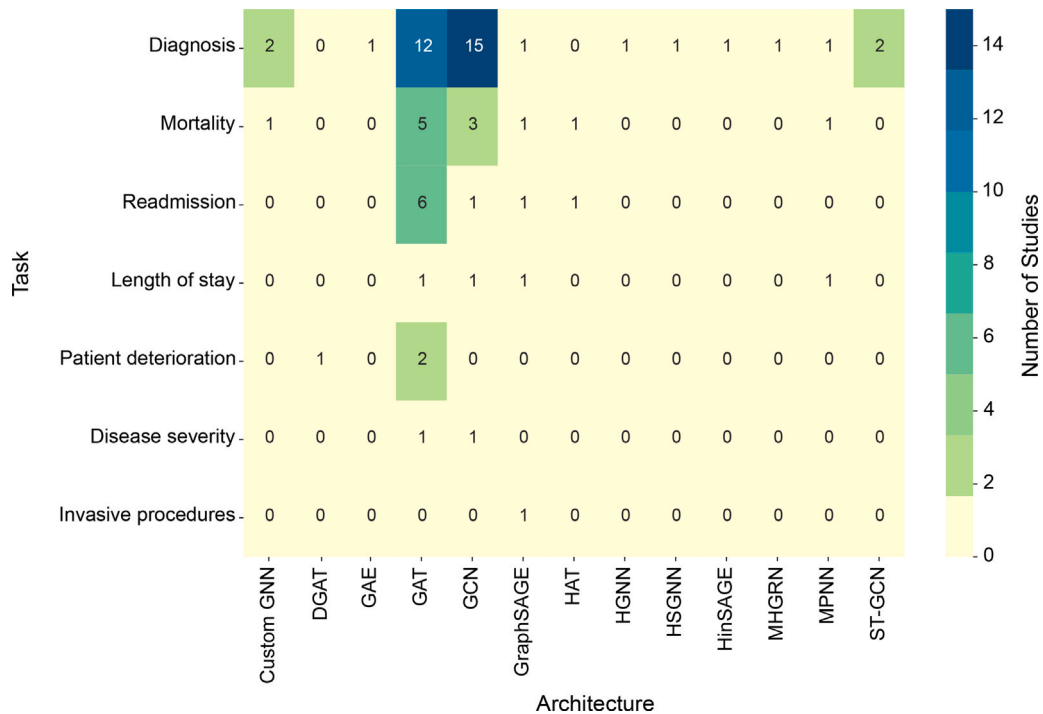


Fig. 6. Heatmap of articles using different GNN architectures for solving different clinical risk prediction tasks. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

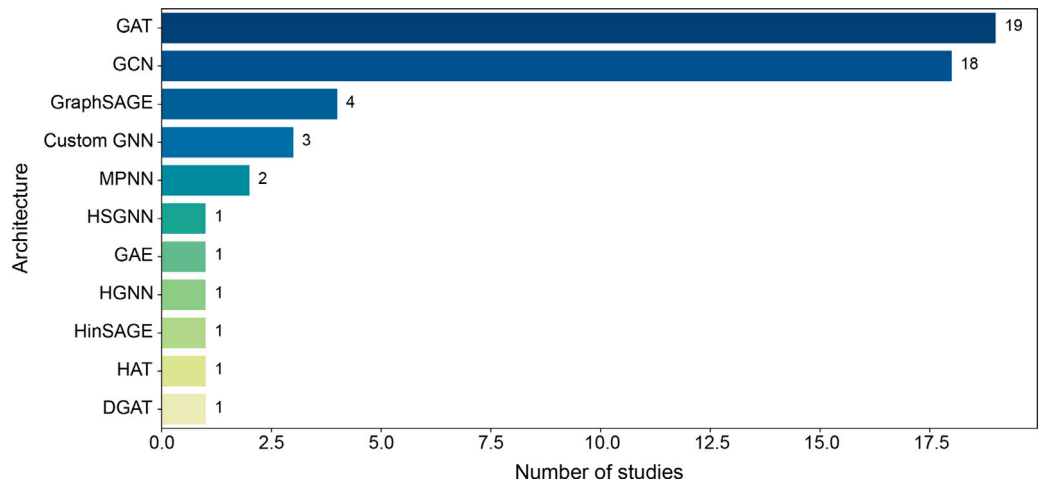


Fig. 7. Distribution of articles using different GNN architectures.

In the evaluated articles, half employed interpretability techniques ($n = 25$, Fig. 8). These methods aim to identify critical nodes, edges, subgraphs, and their features responsible for GNN outputs [120]. The most prevalent approach involved analyzing the t-SNE plot of the embedding space ($n = 10$). The t-SNE method maps high-dimensional data into a lower-dimensional space, preserving local and global information [121]. The resultant clusters, ideally non-overlapping, represent different subtypes learned by the model. Other dimensionality reduction techniques, such as UMAP and PCA, were also used for similar purposes. The second most common approach was the analysis of learned attention weights ($n = 8$). Graph attention layers use these weights to compute attention-guided embeddings for nodes, edges, subgraphs, or combinations [74]. Their magnitude reflects the importance of a given node j 's features to node i , which naturally becomes an interpretability resource.

4.2. Graph data representation

4.2.1. Ontology-based approaches

The primary aim of representation learning for EHR is to transform the input data into an expressive representation that can enhance downstream clinical prediction tasks [15]. The main goal of utilizing GNNs is to incorporate the existing relations among medical events and entities from EHRs into the learned representations. This process is often achieved through knowledge injection, which introduces *a priori* medical knowledge to guide and enrich deep learning architectures [122,123].

Medical knowledge graphs and ontologies contain rich hierarchical information (such as medical events co-occurrence, and “is the cause of” and “is caused by” relations), which can offer a comprehensive and reliable understanding of how clinical concepts interact. Examples include KnowLife [124], a knowledge graph that integrates unstructured biomedical data; the International Classification of Diseases (ICD)

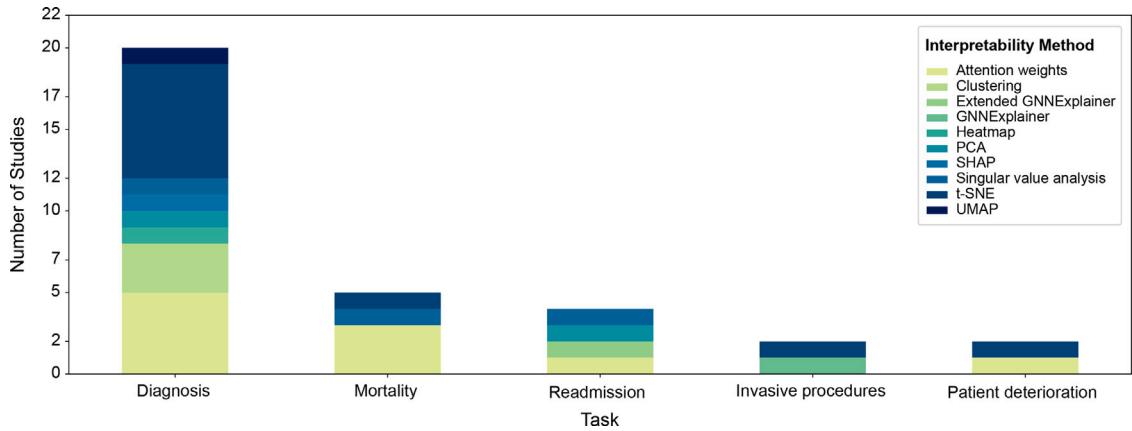


Fig. 8. Distribution of articles using different interpretability techniques. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.).

Table 5
A priori medical knowledge used in GNNs for clinical risk prediction.

Knowledge injection sources	Papers
ICD	[25,28,30,31,80,82,91–93,96,107,108]
CCS	[28,80,92,103,112]
CMeKG	[106,108]
KnowLife	[25]
Other	[17,29,30,105,107,108]

ontology [43], which is the global standard for reporting and recording diseases, symptoms, signs, and other events in all realms of healthcare; the Clinical Classifications Software (CCS), which clusters the numerous individual ICD codes into a smaller number of clinically meaningful categories [125], and CMeKG, the first Chinese medical knowledge graph [126]. All of them and more have been used for domain knowledge injection in GNNs for clinical risk prediction (n = 26); a summary of the main sources can be found in Table 5.

The hierarchical relations in medical ontologies can be naturally represented as parent–child graphs, in which the non-leaf nodes in the tree represent broader, more general classes of medical concepts, and leaf nodes represent more specific instances of these classes [80]. In ICD-9, for example, parental codes 390–459, referring to “Diseases of the Circulatory System”, represent a broad category encompassing various circulatory system conditions. They are connected to subclasses that detail the conditions, such as codes 401–405 “Hypertensive Disease” [127].

A frequently observed method of inputting knowledge in GNNs involves extracting medical codes from a patient’s EHR, especially ICD codes related to diagnoses, and combining them with the graph structure features of hierarchical medical ontology graphs (n = 12, Table 5) [1]. More specifically, the node embeddings in the ontology-based GNN models are learned as a combination of embeddings of the observed medical codes for a patient and the code’s ancestors on the medical graph [11]. This method often depicts each patient’s visit or medical history as a graph with medical codes as nodes [25,103]. Edges indicate mainly the hierarchical relationships among the codes observed in an ontology, and can also be combined with co-occurrence information, indicating comorbidity or a significant relationship between the observed diseases [80].

Some examples of this approach include the Graph Neural networks based Diagnosis Prediction (GNDP) model, which leverages the ICD ontology to predict medical codes occurring at the next visit [28]; and Sherbet, which utilizes hyperbolic embeddings to reconstruct the disease hierarchical structure and predict temporal health

events [31]. MedPath [30] introduces the concept of extracting personalized knowledge subgraphs from ontology graphs for individual patients, which is further explored in MedML [105]. In the Graph-based Structural Knowledge-aware Network (GSKN) model, subgraphs are also utilized, but with the intent to capture deep-level knowledge graph structure information and dynamic representations of medical entities [108]. The HyperGraph-based disease prediction model (EHR2HG) incorporates the ICD hierarchy and hypergraphs to consider higher-order relations among diseases and patients [82]. The Graph ATtention-Embedded Topic Model (GAT-ETM), simultaneously learns node embeddings based on both the ICD (diseases) and ATC (medications) ontology medical codes with a GNN [107].

Other sources of knowledge were also proposed to capture medical event connections that might exist outside the scope of ontologies. In [16], Choi et al. introduced a normalized conditional probability matrix P, which restricts the model’s search space based on the co-occurrence statistics of medical codes in an EHR dataset. In the Joint Medical Ontology Representation Learning (JMRL) [25] and DUAL-GRaPh Representation Learning (DUGRA) [82] models, the information contained in both medical ontologies and co-occurrence statistics of medical codes is explored. Variational regularization has also been used to impose constraints on the learning process and perform structure learning without predefined guiding graphs [18]. Direct medical expert validation has also been employed [105].

4.2.2. Heterogeneous approaches

Recently, new approaches have been leveraging more of the rich multimodal information contained in health records (n = 9, Table 3). Instead of focusing only on hierarchical medical information and structured knowledge graphs, medical events and entities can be represented within a heterogeneous EHR graph. This results in a dense network of interactions that more effectively capture the complexity of a patient’s health status, and can be applied to broader clinical settings and data resources [79,84,110]. While traditional GNN architectures were designed for homogeneous graphs, these new methods employ novel GNNs designed for graphs with often multiple types of nodes and edges. Alternatively, some of the evaluated methods aim to transform a heterogeneous EHR graph into a homogeneous one, allowing it to be processed by standard GNN layers [29].

In Med2Meta [84], Chowdhury et al. uses graph autoencoders to learn feature-specific embeddings for singular medical concept categories in the EHR: demographics, laboratory results, as well as clinical notes. These are then combined into meta-embeddings for downstream tasks, which significantly benefits predicting diagnosis in a patient’s subsequent visit. The HarmOnized Representation learning on Dynamic EHR graphs (HORDE) model generates harmonized medical entity embeddings based on a multimodal dynamic EHR graph [79]. The

approach focuses on two types of nodes: time-invariant nodes with static properties, such as events (diagnoses, procedures, laboratory results) and medical concepts (from unstructured information from clinical notes), and time-varying nodes with dynamic properties, such as patients. Multimodality resulted in a more robust model for disease classification. To minimize the impact of noise in the heterogeneous EHR graph, the Heterogeneous Similarity Graph Neural Network (HS-GNN) [29] model employs a preprocessing method that splits the heterogeneous EHR graph into several homogeneous subgraphs, which are then merged into a single homogeneous graph. The unified graph can then be fed into a GNN for diagnosis prediction.

To handle multi-dimensional time-series EHR data, the Time-aware Context-Gated Graph Attention Network (T-ContextGGAN) [95] model also utilizes a heterogeneous EHR graph, where patient nodes are connected to clinical event nodes such as lab tests, infusion drugs, and prescriptions. The model then uses meta-paths to connect nodes from various time steps for survival prediction. ME2Vec also addresses both the heterogeneity of EHRs and time series data, employing a hierarchical framework to embed medical services, doctors, and patients [85]. Medical services are embedded using a random-walk approach to account for irregular time intervals, while a GNN and a proximity-preserving network embedding approach is used for doctors and patients.

Cho et al. developed a novel EHR graph-database approach using a multi-attributed and multi-relational bipartite graph to represent patients and their relations with hospital visits and clinical events [99]. Then, HinSAGE was employed for predicting cardiovascular disease events, demonstrating superior performance over traditional ML algorithms in a link prediction task.

4.3. Temporality

Given that clinical risk prediction tasks can benefit from considering the temporal aspects of patient records, EHR data are often not represented as a single graph but as a series of graphs observed in different timesteps for a particular patient; or as a dynamic graph, where new EHR information can be added and thus its nodes and edges change over time [79,80,102]

To handle the sequential nature of hospital encounters, several models couple GNNs with recurrent neural networks (RNNs) such as long short-term memory (LSTM) networks or gated recurrent units (GRU) ($n = 18$, Supplementary Material). These architectures operate by passing hidden state information from one input unit to the next, encoding information about the entire sequence of medical events [1]. When coupled with GNNs, this pairing enables the model to leverage both structural knowledge from medical graphs and time dependency between clinical events [23]. For example, in JMRL, an attentive GRU is used to aggregate temporal information between visits represented as graphs for next-visit diagnosis prediction [25]. In the Longitudinal and Graph Integrated (LIGHTED) model, Dong et al. account for the time between visits by concatenating the embeddings of nodes representing patient visits with the raw features observed for that visit and feeding it sequentially into an LSTM [19].

Another approach focused on handling irregular sampling in patient time information is using ordinary differential equations (ODEs). In the Graph Attention and RNN-based Neural Ordinary Differential Equations Model (GROM) model [92], the ODE represents dynamic clinic data as a continuous trajectory influenced by local initial states and the global dynamics of the entire time series, and a time-invariant neural network function determines the whole of the latent trajectory. The data is then fed back to a bidirectional RNN layer to mitigate gradient and sequence length-related issues.

Other studies have handled event sequentiality by employing spatiotemporal GCNs to model both the graph structure and temporality of the EHR data ($n = 2$, Supplementary Material). In these models, temporal information was leveraged by stacking node attributes along the

timestamp and employing a convolution operation to extract features in the temporal domain [28,80].

A dynamic graph, on the other hand, can be characterized as a pair (G, O) , where G refers to a static graph that represents the initial state of the dynamic graph, and O represents a tuple consisting of the event type, the specific event (e.g., edge or node addition or deletion), and its timestamp [128]. In the HORDE model [79], medical events, clinical notes, and patients are represented as nodes in a dynamic graph, with edges indicating event co-occurrence. As patient condition changes over time, LSTM captures these changes, providing a temporal context to the evolving graph structure. [12] adds a new timestamped node in the patient's EHR graph for each registered event. In TAG-Net [83], time-series EHR data is also represented as a dynamic graph to depict a single patient's physiological condition across visits. A GRU model is then employed to create the different representations for each timestamp.

Lu et al. introduced a global dynamic disease graph, shifting the focus from individual patients [102]. They extract subgraphs from observed diseases during patient visits, which are then processed with GRUs to discern temporal disease patterns and forecast potential future outbreaks.

4.4. Clinical risk prediction tasks

4.4.1. Diagnosis prediction

Diagnosis prediction aims to forecast whether a patient will be diagnosed with a medical condition ($n = 36$, Fig. 5). The reviewed studies primarily focused on predicting diagnoses within the current or upcoming visit [16,102]. Still, some extended it to many hours before manifestation or even up to a year or two in advance [18,98]. Most studies have treated the task as a multilabel classification problem ($n = 24$, Supplementary Table). In this context, various probabilities are determined for multiple diseases, allowing a patient to be associated with several diseases.

Graph convolutional networks (GCNs) are frequently employed for diagnosis prediction ($n = 15$; Fig. 6). In [86,129], GCNs were used to learn medical concept embeddings based on a medical ontology. The HealGCN model utilizes a heterogeneous GCN and a symptom retrieval system to enable online disease self-diagnosis [88]. In [28,80], spatiotemporal GCNs were applied to capture the sequential patterns in patient records and predict multiple diseases.

Graph attention networks (GATs) have also been widely utilized for diagnosis prediction ($n = 12$; Fig. 6). They assign different attention weights to specific medical entity nodes, allowing the models to learn which nodes are more relevant for a given prediction task [74]. For example, Choi et al. introduced the Graph Convolutional Transformer (GCT) with an attention mechanism to learn the graphical representation of a patient's visit, highlighting the most significant medical events for future diagnosis prediction [16]. The JMRL model uses GATs with a feedback strategy, incorporating medical knowledge graph embeddings and medical concept co-occurrence to predict multiple diagnoses [25]. The MedPath model utilizes attention weights to provide explanations for medical paths used in diagnosis prediction [30], and the GTGAT model employs a gated tree-based GAT with hierarchical and semantic-aware attention to distill valuable information from nodes, enhancing personalized disease diagnosis performance [98].

Many studies have focused on predicting specific diagnoses ($n = 14$, Supplementary Table), particularly cardiovascular diseases — which can be explained by the high prevalence of positive diagnosis cases in EHR datasets ($n = 5$, Supplementary Table) [130]. For instance, the HinSAGE model was utilized in [99] to predict cardiovascular disease outcomes, whereas [12,84,95] employed GNNs for heart failure prediction. Other specific diagnoses include diabetes, chronic hepatitis B, opioid overdose, lymphocytic leukemia, sepsis, and pediatric COVID, as seen in [19,27,85,94,95,105,110,114]. Furthermore, efforts to diagnose rare diseases have also been observed [17].

4.4.2. Mortality prediction

Mortality prediction aims to determine whether a specific patient will pass away ($n = 10$, Fig. 5). This can be based on the patient's current visit or a particular period after admission or discharge from the hospital or the Intensive Care Unit (ICU) [18,110]. In most cases, mortality prediction has been evaluated along with other clinical tasks such as readmission and diagnosis prediction ($n = 5$, Table 2) [18,26,109]. Furthermore, it is often formulated as a binary classification problem, where 1 predicts expiration, and 0 predicts survival.

Various strategies have been employed to address this problem. In the Temporal Aware Graph Convolution Network (TAGNet) model, EHR time-series data are represented as evolving graphs, and a GCN is used to mine the structural information for mortality forecasting [83]. Rocheteau et al. predict in-hospital mortality using a patient similarity graph [26]. Multimodal information, such as clinical notes and patient correlation, is integrated into a multi-view approach in the Patient Multi-view Multi-modal Feature Fusion Network (PM2F2N) model [104]. Furthermore, AttenSurv aimed to predict patient survival using a global attention mechanism and a GNN to extract and identify latent correlations between clinical risk factors [87].

4.4.3. Readmission prediction

Readmission prediction involves forecasting whether a particular patient will be readmitted to the hospital after discharge or to the Intensive Care Unit (ICU) during the same hospital visit [89,115] ($n = 8$, Fig. 5). This task is often formulated as a binary classification problem, where 1 predicts readmission, and 0 predicts no readmission.

Choi et al. and Wu et al. [16,85] proposed predicting ICU readmissions using a single graph-structured encounter. In [18], readmission at discharge was predicted using a variationally regularized encoder-decoder graph network. [109] utilized hypergraph contrastive learning to predict readmission using patient data collected within the initial 24 h of admission to the ICU [109].

Recent studies have focused on extending the readmission prediction horizon to 30 days. Golmaei and Luo proposed a model that integrates clinical note information with the topological structure of patient networks [89]. Furthermore, Tang et al. utilized a multimodal spatiotemporal GNN incorporating patient similarity to improve readmission predictions [115].

5. Discussion

This section examines the main aspects and perspectives of using GNNs for EHR analysis, returning to the topics mentioned in the Results section.

5.1. Overview

Going beyond the limitations of tabular patient data, GNNs have the differential of leveraging medical events and entity dependencies into predictions to generate more reliable and personalized patient results. Some of these include the use of medical ontology hierarchies as an *a priori* knowledge source, as seen in [25,28,80] (Table 5); patient similarity graphs, as seen in [26,98]; and multi-view approaches, which incorporate the structure of different medical graphs into predictions such as treatment, medication, and diagnosis interactions, as observed in [91,96,104]. A summary of identified EHR graph representations was provided in Table 3, and further details can be found in the Supplementary Material.

5.2. Multimodality

EHR data present inherent complexity owing to their multimodal nature. It encompasses diverse data types, from images to continuous and discrete attributes, including medical images, clinical notes, lab results, and medications. A number of studies have primarily focused on using codes from medical ontologies to represent a patient (Supplementary Table). Such an approach is limited in capturing the rich diversity of information contained within medical records [1]. By integrating various data modalities, a more comprehensive view of a patient's health status can be achieved, thereby enhancing the performance of clinical risk prediction.

It was possible to observe that GNNs can help deal with the challenges posed by this dense amount of patient heterogeneous data, especially when considering the most recent models. For example, in the DeepNote-GNN model, a natural language processing BERT module was also used to leverage clinical notes for readmission prediction. Moreover, [116] used EHRs and genetic reports to predict cancer. A single study, [115], employed imaging data, specifically chest radiographs. Some further methods of accomplishing this include employing multi-view approaches and heterogeneous graphs, as in previously described models. Coupled with adequate GNN architectures, these representations allow the exploration of various relevant relations within multi-modal patient data. For instance, one model segment may focus on learning disease-disease relations while another can emphasize patient similarity. When coupled, they can become an important and holistic resource for modeling the complex topology of EHR relations within patient information.

Furthermore, beyond the data types already present in most EHRs, the integration of novel information from omics data (e.g., genomics and transcriptomics) (only one observed study, [116]) and sensor/wearable data (no observed studies) has the potential to provide additional insights, enriching even further a patient's profile.

5.3. Model evaluation

All studies provided some sort of GNN comparison against other machine learning and deep learning models. For example, [98] compared a Gated Tree-based Graph Attention Network with a multilayer perceptron (MLP) and achieved an improvement in accuracy of almost 9%; and MERGE [101] had an increase in the AUROC compared with other deep learning baselines of almost 16%. It is worth mentioning that the degree of improvement heavily depends on the conditions of the studies, including the model architecture, data resources, and evaluated clinical risk prediction tasks.

Furthermore, EHR datasets are predominantly organized in tabular format. This allows researchers to design different EHR graph structures optimized for various downstream tasks. However, this flexibility also challenges the comparability of graph-based models. For example, even if two GNNs are trained on an identical EHR dataset (such as observed: 23 of the 50 articles used a benchmark dataset, MIMIC-III, for model deployment; Table 4), the model results may be complex to compare owing to the differences in the GNN architectures and the unique EHR graph topologies each one of them proposes. Furthermore, these models may interpret tasks differently. For instance, models might vary in defining the threshold for assigning a '1' (indicating positive) or '0' (indicating negative) in clinical risk prediction tasks.

This complexity shows that making predictions in the medical field is still a delicate task that depends heavily on the available resources and the limitations of the architectures of the models. Moreover, model variability underscores the need for unified assessment metrics and cross-model interpretability tools to improve the comparability of the GNN models. This is especially crucial for clinical tasks, as differences in model predictions, whether from varying graphical structures or interpretive paradigms, can lead to different critical clinical interventions, directly impacting a patient's health.

5.4. Interpretability

Regarding interpretability, it was observed that half of the models employed some form of interpretability technique (Fig. 8). Among these, highlights include attention weights and visualizations of the embedding space with dimensionality reduction techniques. For example, in [26], the attention weights explain the model's behavior in learning to correctly diagnose a patient based on assigning higher weights to other patients with shared diagnoses. In [86,90], the embedding space was visualized, providing insights into the learned clinical clusters.

However, these interpretability resources may only partially explain the models, especially if they involve high architectural and preprocessing complexity. Deep learning methods are often referred to as 'black boxes' because they operate by inputting complex data, learning abstract patterns, and producing difficult-to-interpret results, omitting their internal logic to users [131]. Given this, overall, the more heterogeneous the data used to represent patients, the more complex and challenging it is to understand the GNN models, making it a double-edged sword. For example, a GNN based only on diagnosis codes can be less complex to examine than one based on multiple medical relations and entities. The lack of graphical ground-truth explanations makes interpretability even more complex, especially since most EHR datasets are tabular, and standard evaluation strategies for graph machine learning are still emerging [132]. In this sense, it is crucial to move beyond traditional interpretability methods to ensure the practical utility of GNN-based systems in the medical field, given that predicting health outcomes is a critical subject that requires thorough understanding to be implemented in real clinical settings.

Although the identified techniques provide a level of insight, efforts should be directed towards incorporating novel GNN interpretability approaches, visual analytics, and user experience (UX) methods, enabling medical professionals to effectively evaluate the outputs of GNNs and provide feedback on the system's predictions. Some recently proposed methods include GNN-Explainer [133] and neuron analysis [134]. More possibilities are described in [120,135] and point to relevant directions for investigation, especially when combined to include medical professionals in the model development process.

5.5. Temporality

Most existing GNN models applied to EHR data focus primarily on capturing the sequential nature of events, disregarding the irregular time aspect of clinical records. Data irregularity has already been pointed out as a major challenge in modeling patient temporal data [7]. Models often overlook the exact time gaps between medical events and visits; for example, if a visit happened 1-day apart or one year apart. This is a crucial gap, as the time intervals between clinical events can carry important information about the progression of a patient's health condition. By considering irregular time intervals between events, GNN models can offer a more comprehensive representation of temporal dynamics and potentially enable more accurate predictions.

Using GNNs that can handle time series information and coupling GNNs with other deep learning architectures that can leverage irregular sequentiality represent interesting investigation directions. For example, GNN architectures that can handle dynamic EHR graphs, as they can update the graph structure as new timestamped patient data comes in [79]; and modules based on ordinary differential equations (ODEs), well-suited for modeling continuous-time data [92]. Other GNN techniques that can be used for time series processing are listed in [136].

5.6. Prediction tasks

Among the evaluated articles, diagnosis prediction was the predominant task, as it benefits from abundant and standardized diagnostic code information in EHRs ($n = 36$, Fig. 5). It is essential to predict diagnoses accurately, but clinical tasks beyond diagnosis prediction also need to be diversified. Recent research has focused on novel topics, such as survival analysis, hospitalization risk, patient deterioration, and disease severity (Fig. 5). It was also observed a need for more diversity in GNN models evaluating other clinical tasks, with significantly fewer published articles than diagnosis prediction (Fig. 6). Also, models for predicting rare diseases and conditions are incentivized, as among the analyzed studies, only one aimed to tackle this problem [17]. The high concentration and good performance of GNN architectures tested for diagnosis prediction suggest that other prediction tasks can also benefit from these techniques. Moreover, there is still ample room for further exploration in evaluating these clinical risk tasks with better patient representation based on graphs, especially those based on a more heterogeneous, comprehensive view of a patient's health state.

Finally, clinical tasks primarily aim at predicting outcomes related to health concerns. Although reactive medicine is vital for patient care, exploring the favorable factors that enhance patient health or optimize treatment processes presents valuable opportunities for a more holistic and patient-centered approach. In contrast to reactive medicine, proactive and personalized medicine dedicates more time and resources to disease prevention, early diagnosis, and treatment at stages when it is more cost-effective and potent. This approach also emphasizes managing chronic conditions before they escalate and lead to severe complications.

5.7. Limitations

This study did not cover certain aspects, providing opportunities for future research. This includes the evaluation of GNN model bias and fairness across diverse patient populations, an area increasingly recognized for its importance in clinical settings, as well as patient data confidentiality. Understanding how these models perform across different patient demographics and how to handle patient data safely in GNN models is critical in clinical settings to ensure equitable health outcomes.

It is also important to highlight the need to investigate the integration of these models into actual clinical workflows. The direct application and evaluation of these models in clinical environments, alongside a thorough comparative analysis with non-ML models like clinical expert systems, were outside the study's focus but are valuable directions for future research. This involves considering how these models interact with and augment the decision-making processes of healthcare professionals.

Additionally, the scalability of these models, especially in terms of computational requirements, was not a primary focus of this study. GNNs, mainly when dealing with large and complex datasets typical in healthcare, can require significant computational power. Investigating ways to optimize these models for more widespread and cost-effective use in diverse clinical environments is essential for subsequent studies.

In regard to the narrative review approach, it proves beneficial for a broad and comprehensive overview. Yet, it presents some limitations. Due to the rapidly evolving nature of GNN applications in clinical risk prediction, some recent developments may not have been included. In addition, narrative reviews can limit the generalizability of findings and thus lead to potential bias in the selection and interpretation of studies. These limitations highlight the need for future systematic reviews and meta-analyses to enhance the fundamental insights provided by this study.

Finally, this study specifically focused on studies that employed GNN approaches. Given the novelty of this field and the often inconsistent terminology, there is a possibility of unintentionally omitting specific works. However, the filtering criteria were carefully designed to maximize the inclusion of relevant studies, ensuring a comprehensive listing despite the potential for omissions.

6. Conclusion

The comprehensive review presented in this paper examined the application of GNNs for clinical risk prediction using EHRs. Initially, a background on EHR graph representation learning was provided, along with its relevance to relational medical data, and an introduction to GNNs and graph analysis. Next, the paper highlighted state-of-the-art GNN approaches that are effective in modeling EHR data, including the graph convolutional network (GCN), graph attention network (GAT), and graph autoencoder (GAE).

The Results section provided statistics on the analyzed articles and detailed the analysis around three axes: data representation, temporality, and clinical prediction tasks. It was possible to identify a growing trend of articles focusing on this topic, and the positive outcomes of the analyzed models demonstrate the significance and potential of the area. Among the architectures employed, GAT and GCN were the most common, and the task on which these models focused the most was the prediction of diagnoses, followed by mortality and readmission. Furthermore, there was a predominance of the MIMIC-III dataset as a resource for building the models.

Finally, the Discussion section presented open challenges in the area. Future research directions include developing models that effectively handle multimodal, heterogeneous, and irregular time information in the EHR data. Additionally, interpretability should be emphasized, especially considering enhancing clinicians' understanding of the predictions, and efforts should be directed towards diversifying the scope of investigated clinical risk prediction tasks. These perspectives will contribute to more informed healthcare decision-making and ultimately improve patient care.

CRediT authorship contribution statement

Heloísa Oss Boll: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ali Amirahmadi:** Writing – review & editing, Conceptualization. **Mirfarid Musavian Ghazani:** Writing – review & editing, Conceptualization. **Wagner Ourique de Moraes:** Writing – review & editing, Conceptualization. **Edison Pignaton de Freitas:** Writing – review & editing, Conceptualization. **Amira Soliman:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Farzaneh Etminani:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Stefan Byttner:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Mariana Recamonde-Mendoza:** Writing – review & editing, Visualization, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jbi.2024.104616>.

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